Gbirison Iyimide Gabriel

VUG/SEN/24/12154

ENTITIES AND ATTRIBUTES:

ENTITY	ATTRIBUTES	PRIMARY KEY
Department	DepartmentName	DepartmentName
Student	MatricNo, name	MatricNo
Lecturer	name	name
Course	courseCode, courseTitle	courseCode
Enrolement	session	

UNNORMALIZED REGISTRATION SHEET:

matric_no	student_name	student_dept	course_codes	course_titles	lecturers	lecturer_depts	session
20231234	Ada Obi	Software Engineering	SEN307, SEN301	Database Systems, Software Architecture	Dr A. Okafor, Dr M. Bello	Software Eng., Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307, COS301	Database Systems, Operating Systems	Dr A. Okafor, Mrs F. Eze	Software Eng., Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

1NF: Lists were splitted and individual rows were created.

Matric_no	Student_name	Student_dept	Course_codes	Course_titles	lecturers	Lecturer_depts	session
20231234	Ada Obi	Software Engineering	SEN 307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231234	Ada Obi	Software Engineering	SEN 301	Software Architecture	Dr M. Bello	Software Engineering	2026/2027
20231235	John Musa	Computer Science	SEN 307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231235	John Musa	Computer Science	COS 301	Operating Systems	Mrs F. Eze	Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN 301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

2NF: Columns were moved and new tables created.

Table 1: Student

Matric_no (PK)	Student_name	Student_dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

Table 2: Course

Course_code (PK)	**Course_title**	**lecturer**	**Lecturer_dept**
SEN 307	Database Systems	Dr A. Okafor	Software Engineering
SEN 301	Software Architecture	Dr M. Bello	Software Engineering
COS 301	Operating Systems	Mrs F. Eze	Computer Science

Table 3: Enrolment

matric_no (PK, FK)	Course_code(PK, FK)	**session**
20231234	SEN 307	2026/2027
20231234	SEN 301	2026/2027
20231235	SEN 307	2026/2027
20231235	COS 301	2026/2027
20231236	SEN 301	2026/2027

3NF: Columns that described other columns were moved, and new tables were created.

Table 1: Department

Dept_id(PK)	Dept_name
1	Software Engineering
2	Computer Science

Table 2: Student

Matric_no(PK)	Name	Dept_id(FK)
20231234	Ada Obi	1
20231235	John Musa	2
20231236	Grace Okon	1

Table 3: Course

Course_code (PK)	Course_title
SEN 307	Database Systems
SEN 301	Software Architecture
COS 301	Operating Systems

Table 4: Lecturer

Lecturer_id(PK)	Lecturer_name	Dept_id(FK)
1	Dr A. Okafor	1
2	Dr M. Bello	1
3	Mrs F. Eze	2

Table 5: Enrolment

Matric_no(PK, FK)	Course_code(PK, FK)	Session
20231234	SEN 307	2026/2027
20231234	SEN 301	2026/2027
20231235	SEN 307	2026/2027
20231235	COS 301	2026/2027
20231236	SEN 301	2026/2027

Table 6: CourseAndLecturer

Course_code(PK, FK)	Lecturer_id(PK, FK)	Session(PK)
SEN 307	1	2026/2027
SEN 301	2	2026/2027
COS 301	3	2026/2027

FINAL SCHEMA:

1. Department (dept_id PK, dept_name)
2. Student (matric_no PK, name, dept_id FK -> Department)
3. Lecturer (lecturer_id PK, lecturer_name, dept_id FK -> Department)
4. Course (course_code PK, course_title)
5. Enrolment (matric_no PK, FK -> Student; course_code PK, FK -> Course; session)
6. CourseAndLecturer (course_code PK, FK -> Course; lecturer_id PK, FK -> Lecturer; session PK)